DEPARTMENT OF COMPUTER & INFORMATION SYSTEMS ENGINEERING
BACHELORS IN COMPUTER SYSTEMS ENGINEERING
Course Code: CS-327
Course Title: Computer Communication Networks
Complex Engineering Problem
Fall Semester 2026
Grading Rubric
TERM PROJECT

Project: Design, Implementation and Evaluation of a Multi-Host Networked Application

Group No.: _____ Application Option Selected (A to E): _____

Group Members:

Student No.	Name	Roll No.
S1		
S2		
S3		
S4		

CRITERIA AND SCALES				Marks Obtained			
				S1	S2	S3	S4
Criterion 1: Protocol Design and Layering Analysis (CLO-1) [10 marks]							
1 (0 to 2)	**2 (3 to 5)**	**3 (5 to 7)**	**4 (8 to 10)**				
No formal protocol specification is provided. Message formats exist only inside the code. Packet capture is absent, or a screenshot is included with no explanation.	Message types are listed, but the message format, the order of exchange or the action taken on receipt is incomplete. A capture is included but the protocol layers are not correctly identified.	A usable specification covering message formats, legal exchange order and action on receipt. The capture correctly identifies the frame, the datagram, the segment and the application message.	A precise specification that another group could implement from, including status and error codes. The encapsulation analysis relates each captured header to the layered model and explains what that layer contributed.				
Criterion 2: Implementation and Multi-Host Deployment (CLO-2) [10 marks]							
1 (0 to 2)	**2 (3 to 5)**	**3 (5 to 7)**	**4 (8 to 10)**				
Code is incomplete or does not run. All components execute on a single machine using 127.0.0.1. No evidence of host IP configuration. Assessed as an incomplete implementation.	The application runs, but on fewer than three independent hosts, or the deployment evidence such as IP configuration and network diagram is missing or does not match what is demonstrated.	The application runs correctly across three independent hosts holding distinct IP addresses. The network diagram and IP addressing plan are present and consistent with the live demonstration.	As level 3, and the system handles concurrent clients or multiple servers cleanly. Sockets are used directly with no application protocol library, and the demonstration runs without intervention.				

CRITERIA AND SCALES				Marks Obtained			
				S1	S2	S3	S4
Criterion 3: Measurement, Analysis and Engineering Justification (CLO-2) [10 marks]							
1 (0 to 2)	**2 (3 to 5)**	**3 (5 to 7)**	**4 (8 to 10)**				
No measurements, or figures quoted with no stated method. The transport choice is asserted rather than justified. Design tensions and failure behaviour are not addressed.	Some measurements are taken, but from single runs, without a comparison condition, or the results are presented without explanation. Transport justification is a single line.	Measurements are repeated and averaged across at least two conditions, presented in a table and a graph, and explained using the delay model. Transport choice is justified against the service requirements. At least three design tensions are discussed and failure behaviour is demonstrated.	As level 3, and the analysis is genuinely critical. Anomalies in the data are noticed and explained rather than hidden, trade-offs are resolved by reference to the group's own numbers, and the limitations of the measurement method are acknowledged.				
Criterion 4: Sustainability and Efficiency Analysis (CLO-3) [10 marks]							
1 (0 to 2)	**2 (3 to 5)**	**3 (5 to 7)**	**4 (8 to 10)**				
Absent, or generic statements about green computing with no connection to the project or its data.	Efficiency is mentioned, but nothing is quantified from the group's own measurements.	Quantifies the bytes saved by a specific design choice such as caching, connection reuse or message compactness, using the group's own measurements, and expresses the saving as a percentage.	As level 3, and projects the saving to a realistic population of users, reasons about the bandwidth and energy implications at scale, and states the assumptions behind the projection honestly.				
Criterion 5: Report Quality, Demonstration and Individual Contribution [10 marks]							
1 (0 to 2)	**2 (3 to 5)**	**3 (5 to 7)**	**4 (8 to 10)**				
The report does not follow the required structure, exceeds the page limit, or omits key sections. One or more members are absent from the demonstration, or cannot explain the work.	The report broadly follows the structure but sections are thin or poorly presented. The demonstration needs repeated prompting. The contribution table is missing or unsigned.	The report follows the required structure within the page limit, is clearly written with labelled figures and tables, and the demonstration is fluent. The contribution table is complete and signed by all members.	As level 3, and every member can answer questions on any part of the design, not only their own. The report reads as one coherent document rather than stitched sections, and the code runs as submitted.				

Final Score = Criterion 1 + Criterion 2 + Criterion 3 + Criterion 4 + Criterion 5

= _____ out of 50